

TU VU

 [tuvllms.github.io](https://github.com/tuvllms) |  tuvu@vt.edu |  [Google Scholar](#) |  [@tuvllms](#) |  [@tuvllms](#)

APPOINTMENTS

Virginia Tech

Fall 2024 — *present*

Assistant Professor, Computer Science

Research Interests: *large language models & natural language processing*

Google

Faculty Researcher

Spring 2025 — *present*

Google DeepMind Research Scientist

2023 — 2024

EDUCATION

University of Massachusetts, Amherst

2016 — 2023

M.S/PH.D. in Computer Science

Advisor: [Mohit Iyyer](#)

2018 — 2023

Thesis committee: [Mohit Iyyer](#), [Subhransu Maji](#), [Hamed Zamani](#), [Thang Luong](#), [Colin Raffel](#)

Vietnam National University, Hanoi

2009 — 2013

B.S. Honors Program in Computer Science

Highest distinction (class rank: 1st)

PROFESSIONAL EXPERIENCE

Google DeepMind

Fall 2022 — Spring 2023

Research Intern *with* [Thang Luong](#) & [Quoc Le](#)

Google DeepMind

Summer 2022

Research Intern *with* [Thibault Sellam](#) & [Elizabeth Clark](#)

Google DeepMind

Winter 2021 — Spring 2022

Research Intern *with* [Noah Constant](#)

Google DeepMind

Summer 2021 — Fall 2021

Research Intern *with* [Daniel Cer](#) & [Noah Constant](#)

Google DeepMind

Winter 2020 — Spring 2021

Research Intern *with* [Thang Luong](#) & [Quoc Le](#)

Google DeepMind

Summer 2020

Research Intern *with* [Grady Simon](#) & [Zi Yang](#) & [Nan Hua](#)

Microsoft Research

Summer 2019

Research Intern *with* [Tong Wang](#) & [Tsendsuren Munkhdalai](#) & [Adam Trischler](#)

*: equal contribution. For my latest research papers, see my [Google Scholar](#) profile.

WikiSkill: Compiling Agent Experience into Persistent Knowledge for Skill Evolution

Liyan Tang, Cyrus Rashtchian, Chun-Sung Ferng, Andrew Tomkins, Da-Cheng Juan, **Tu Vu**

Under review @ ICLR 2027

// VentureBeat feature // DAIR.AI's Top AI Papers of the Week

EvoSkill: Automated Skill Discovery for Multi-Agent Systems

Salaheddin Alzubi, Noah Provenzano, Jaydon Bingham, Weiyuan Chen, and **Tu Vu**

COLM 2026

// Pioneering work on skill evolution

SealQA: Raising the Bar for Reasoning in Search-Augmented Language Models

Thinh Pham, Nguyen Nguyen, Pratibha Zunjare, Weiyuan Chen, Yu-Min Tseng, **Tu Vu**

ICLR 2026

// Our benchmark dataset has been used by Google's Gemini, Qwen, DeepSeek, Kimi, and Perplexity

The Master Key Hypothesis: Unlocking Cross-Model Capability Transfer via Linear Subspace Alignment

Rishab Balasubramanian, Pin-Jie Lin, Rituraj Sharma, Mohit Bansal, **Tu Vu**

NeurIPS 2026 Spotlight (Top 1.3% out of 30K+ submissions)

Foundational Autoraters: Taming Large Language Models for Better Automatic Evaluation

Tu Vu*, Kalpesh Krishna*, Salaheddin Alzubi, Chris Tar, Manaal Faruqui, Yun-Hsuan Sung

EMNLP 2024

// The top-performing generative model on RewardBench as of July 15, 2024, trained only on publicly available data

Gemini: A Family of Highly Capable Multimodal Models

Google Gemini Team: Rohan Anil, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew Dai, Anja Hauth, and others including

Tu Vu

Technical report 2023

// Google AI Blog

FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation

Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong

ACL 2024 Findings

// ZDNET feature // DAIR.AI's Top ML Papers of the Week // Our dataset and method have inspired or been used for the development of Google's Gemini, Perplexity.AI's Online LLMs, You.com, and Contextual AI's RAG 2.0

The Flan Collection: Designing Data and Methods for Effective Instruction Tuning

Shayne Longpre, Le Hou, **Tu Vu**, Albert Webson, Hyung Won Chung, Yi Tay, Denny Zhou, Quoc Le, Barret Zoph, Jason Wei, and Adam Roberts

ICML 2023

// Pioneering work on large-scale instruction tuning // Google Research Blog

Mixture-of-experts meets instruction tuning: A winning combination for large language models

Sheng Shen, Le Hou, Yanqi Zhou, Nan Du, Shayne Longpre, Jason Wei, Hyung Won Chung, Barret

Zoph, William Fedus, Xinyun Chen, **Tu Vu**, Yuexin Wu, Wuyang Chen, Albert Webson, Yunxuan Li, Vincent Zhao, Hongkun Yu, Kurt Keutzer, Trevor Darrell, and Denny Zhou

ICLR 2024

// Pioneering work on combining mixture-of-experts and instruction tuning

SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer

Tu Vu, Brian Lester, Noah Constant, Rami Al-Rfou, and Daniel Cer

ACL 2022

// Pioneering work on soft prompt tuning & transfer // Headlines of Google AI's Natural Language Accelerated Newsletter Q1, 2022

Exploring and Predicting Transferability across NLP Tasks

Tu Vu, Tong Wang, Tsendsuren Munkhdalai, Alessandro Sordoni, Adam Trischler, Andrew Mattarella-Micke, Subhransu Maji, and Mohit Iyyer

EMNLP 2020

// Pioneering work on exploring task transferability in the LLM era

RESEARCH AWARDS & FUNDING

As a lead PI, I have raised \$1.2M in research funding from NSF and industry labs

Google Cloud credits (\$15,000)	2026
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PI: Tu Vu

NSF CISE Future Computing Research (Future CoRe) Award (\$821,351)	2026
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PI: Tu Vu, Co-PI: Mohit Iyyer; Personal share: \$421,351

Amazon - VT Faculty Research Award (\$75,000)	2026
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PI: Tu Vu

Commonwealth Cyber Initiative (CCI) Award (\$35,000)	2026
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PI: Bimal Viswanath, Co-PI: Tu Vu; Personal share: \$5,000

Gemini Academic Program Award (\$30,000 Gemini credits)	2026
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PI: Tu Vu, Student: Noah Provenzano

MGI Scaling Scholarship Award (\$4,000)	2026
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PI: Tu Vu

Sentient Labs (\$35,000)	2026
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PI: Tu Vu

Google Research Award (\$50,000)	2025
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PI: Tu Vu

Google DeepMind Research Award (\$30,000)	2025
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PI: Tu Vu

Amazon - VT Faculty Research Award (\$130,000)	2025
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PI: Tu Vu

Google AI Studio Tier 1	2025
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PI: Tu Vu

VT New Faculty Mentoring Grant (\$1,500)	2025
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PI: Tu Vu

ADVISING

PHD ADVISEES:

Mikaela Cankosyan, 1 st year PhD student	Fall 2026 — <i>present</i>
Noah Provenzano, 1 st year PhD student	Fall 2026 — <i>present</i>
Weiyuan Chen, 2 nd year PhD student	Fall 2025 — <i>present</i>
Yu-Min Tseng, 2 nd year PhD student	Fall 2025 — <i>present</i>
Rishab Balasubramanian, 3 rd year PhD student	Fall 2024 — <i>present</i>
Thinh Pham, 3 rd year PhD student	Fall 2024 — <i>present</i>
Quyet Do, 3 rd year PhD student	Fall 2024 — <i>present</i>
Pin-Jie Lin, 3 rd year PhD student	Fall 2024 — <i>present</i>

MASTERS ADVISEE(S):

Rituraj Sharma, 2 nd year MS student	Fall 2025 — <i>present</i>
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UNDERGRAD ADVISEE(S):

Nguyen Nguyen, senior UG student	Spring 2025 — <i>present</i>
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HIGH SCHOOL ADVISEE(S):

Samson Ou, Blacksburg High School	Fall 2026 — <i>present</i>
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ALUMNI:

Jaydon Bingham, senior UG student	Spring 2026
Christian Calvo, senior UG student	Spring 2026

OTHERS:

Zhenting Qi, Student Researcher @ Google	Summer — Fall 2025
Prateek Yadav, Research Intern @ Google DeepMind	Summer 2024 — Spring 2025
Simeng Han, Student Researcher @ Google DeepMind	Summer 2024 — Spring 2025
Salaheddin Alzubi, Masters student @ UMass Amherst	Fall 2022 — Spring 2023
Dheeraj Mekala, PhD student @ UCSD	Spring — Summer 2022

ON COMMITTEE:

Parshin Shojaee, VT CS PhD	2025
Gaurav Srivastava, VT CS MS	2025
Hani Alomari, VT CS PhD	2025 — present
Mridul Khurana, VT CS PhD	2025 — present
Xiaohang Tang, VT CS PhD	2025 — present

TEACHING

CS 5624: Natural Language Processing	Fall 2026
CS 6804: Frontier AI Systems	Spring 2026
CS 4804: Introduction to AI	Fall 2025
CS 5624: Natural Language Processing	Spring 2025

RECENT INVITED TALKS & LECTURES

Transformers Invited lectures @ The New Turing Institute	September 2026
Efficient Model Development through Fine-tuning Transfer Lightning talk @ The Amazon-Virginia Tech AI Workshop	October 2025
Attention Mechanisms and Transformers Invited lectures @ The New Turing Institute	August 2025
SealQA: Raising the Bar for Reasoning in Search-Augmented Language Models Featured invited speaker, Open AGI Symposium @ UC Berkeley	August 2025
Multimodal LLMs & Test-time Scaling Invited lecture @ The New Turing Institute	June 2025
Efficient Model Development in the Era of Large Language Models Qualcomm	November 2024
Efficient Model Development in the Era of Large Language Models Mila / McGill NLP Seminar	October 2024
Efficient Adaptation of Large Language Models Graph Neural Networks Reading Group, Google	November 2023
Effective and Efficient Transfer Learning in the Era of Large Language Models Faculty job talk at multiple institutions	Spring 2023
Overcoming Catastrophic Forgetting in Zero-Shot Cross-Lingual Generation Parameter Efficient Tuning Methods Sync, Google	October 2022
Transfer Learning with Large-scale Language Models Lecture @ The New Turing Institute	August 2022
The Appeal of Parameter-efficient Transfer Learning Natural Language Accelerated Team, Google	June 2022
SPoT: Better Frozen Model Adaptation through Soft Prompt Transfer Parameter Efficient Tuning Methods Sync, Google	December 2021

EXTERNAL SERVICE

Senior Area Chair for NAACL 2027
Action Editor for TMLR

Area Chair for ICLR 2027; ACL 2026, 2025, 2024; EMNLP 2026, 2025, 2024; NAACL 2025; EACL 2027, 2026; COLING 2025

Session Chair (Machine Learning for NLP) @ EMNLP 2024

Program Committee/Reviewer for NATURE 2025, NEURIPS 2025, 2024; ICLR 2026, 2025; ICML 2026, 2025; TL4NLP@NEURIPS 2022; MCDL@ICLR 2025; COLM 2026, 2025, 2024; AAAI 2023; ACL 2022, 2021, 2020, 2019; EMNLP 2022, 2021; NAACL 2022, 2021; COLING 2020; CoNLL 2019; INLG 2020, 2019

INSTITUTIONAL SERVICE

CS@VT Faculty Search Committee	2025 — 2026
CS@VT Graduate Admissions Committee	2025 — 2026

SELECTED MEDIA

WIKISKILL: VentureBeat	2026
VT News: Thinking small: How small language models could lessen the AI energy burden	2026
VT News: Amazon-VT Initiative Faculty Award	2025
GEMINI: Google AI Blog	2023
FRESHLLMs: ZDNET	2023
THE FLAN COLLECTION: Google Research Blog	2023
SPoT: Headlines of Google AI's Natural Language Accelerated Newsletter	Q1, 2022

OTHER AWARDS & HONORS

Google Student Researchships	2020 — 2023
UMass Amherst Graduate Assistantships	2016 — 2023
Honda Y-E-S Award for young engineers and scientists, Vietnam <i>// in the top 10 nationally</i>	2013
Outstanding Academic and Co-curricular Achievements, Vietnam National University	2013
Prominent Young Figure Award, Vietnam National University	2010 & 2012
First Runner-up Prize, International Programming Contest, Japan <i>// ranked 2nd among 64 teams internationally</i>	2011
Outstanding Young Talent of the Capital City, Vietnam <i>// in the top 100 most outstanding young talents selected from a wide range of fields</i>	2010
Champion Prize, National Mathematical Olympiad, Vietnam <i>// ranked 1st among more than 600 contestants nationally</i>	2010
A number of prizes in National/International Olympiads (in both Mathematics and Informatics)	2009 — 2013
A number of academic scholarships for undergraduate students	2009 — 2013

*: original inventor

Frozen Model Adaptation Through Soft Prompt Transfer

Tu Vu*, Daniel Cer, Noah Constant, Brian Lester, Rami Al-Rfou

U.S. Patent Application, 17/863,840

Task Augmentation and Self-training for Improved Few-shot Learning

Thang Luong, **Tu Vu***, Quoc Le, Grady Simon

U.S. Patent Application, 17/826,690